Finance Performance Analysis

A graduate portfolio project using SQL Server and Power BI

This case study explains a finance analysis project for a fictional distributor. The main goal was to compare actual sales results with the budget and find areas that need more attention. In 2025, revenue increased from the previous year, but it was still below budget. Technology had the largest gross profit shortfall.

Project aim

I chose this project to support my interest in an entry level finance data analyst role. It brings together SQL, Power BI and basic financial analysis. The report focuses on three questions: Did revenue meet the budget? Did the business earn the planned gross profit? Which product categories and regions need further review?

Data and tools

The project uses 6,912 synthetic sales records from January 2024 to December 2025. There are three product categories and four regions. The data also includes monthly budgets. All amounts are in US dollars. The records were created for this project and do not belong to a real company.

SQL Server stores and prepares the data. Power Query imports the SQL results into Power BI. DAX measures calculate revenue, gross profit, budget differences and growth from the previous year.

Main results for 2025

Measure	Result
Net revenue	$3,761,180.89
Budget revenue	$3,931,645.14
Revenue below budget	$170,464.25 or 4.3%
Gross profit	$937,548.24
Gross margin	24.9% actual and 29.4% budget
Revenue growth from 2024	7.9%

How the data was prepared

The SQL script creates a separate database called FinancePortfolio2025. It loads sales and budget records into tables and adds rules to check the data. For example, each sale has a unique ID, and a discount cannot be higher than the gross sale amount.

Sales are first grouped by month, product category and region. The monthly sales totals are then joined to the budget. This step is important because one monthly budget should not be repeated for every sale. The final reporting view has 288 rows.

The model uses separate date, product and region tables. Each table has a one to many relationship with the monthly finance data. This allows users to filter the report by year, category and region.

Financial measures

Net revenue is gross revenue minus discounts. Gross profit is net revenue minus the cost of goods sold. Gross margin is gross profit divided by net revenue. I use the total amounts for this calculation because taking an average of row percentages can give a misleading result.

Budget variance means the difference between the actual amount and the budget. A positive revenue or gross profit variance is favourable. A margin difference is shown in percentage points. For example, 24.9% actual margin compared with 29.4% budget margin is about 4.5 percentage points lower.

Power BI report

The first page gives a financial overview. It shows the main measures, monthly revenue against budget, gross profit by category and results by region. The second page gives more detail about budget differences, discount rates and changes in margin.

Both pages include year, category and region filters. The filters are separate on each page, so users should choose the same filters when comparing pages. The saved report starts with 2025. The Power BI file includes imported data, so users can view it without connecting to SQL Server.

Checks on the results

Nine SQL Server checks passed. These checks cover record counts, duplicate reporting rows, missing source combinations, revenue and budget totals, sales counts and the date table. The 2025 Power BI totals were also compared with the SQL results. Both report pages were opened and checked in Power BI Desktop.

What the results show

Revenue grew by 7.9% compared with 2024. However, it was about $170,464 below the 2025 budget. This shows why it is useful to compare results with both the previous year and the budget. Sales can grow while the business still misses its plan.

Technology had the largest gross profit shortfall at about $168,288. This was most of the total gross profit shortfall of about $220,103. Technology also had a discount rate of 9.0%, while the other categories were about 2.5%. These results suggest that discounting and costs need closer review.

West had the largest regional revenue shortfall at about $93,162. A useful next step would be to review Technology sales in West and compare them with the plan. The report helps identify where to look, but more information would be needed to understand the business reasons.

Suggested actions

I would suggest reviewing the approval process for Technology discounts and checking whether supplier costs have changed. I would also review the sales plan for West. The business could then track revenue and gross profit against budget each month to see whether results improve. These are suggested actions, and no financial savings have been measured.

Limitations

This is a learning project using synthetic data. The data was designed to include higher Technology discounts and costs, and weaker Technology sales in West. The findings show how the analysis works. They are not evidence about a real business.

Gross profit does not include operating expenses, tax or interest. The report therefore does not show net profit or cash flow. The reporting data is monthly, so it should not be used for daily analysis. Growth measures should be read with one year selected. There is no 2023 data for a 2024 comparison.

Learning focus and project link

The main learning areas are joining data at the correct level, checking totals, building a clear model and explaining financial results in simple words. AI tools helped create the data, code, report and documentation. This project is part of my learning, rather than paid work or a real business project.

The SQL files, Power BI report, results preview and supporting notes are available on GitHub:

https://github.com/james52p/finance-performance-sql-powerbi